

ADITYA KAKI

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EDUCATION

Virginia Polytechnic Institute and State University (Virginia Tech)

Blacksburg, VA

B.S. Computer Engineering, Machine Learning Concentration

Expected May 2027

- **Relevant Coursework:** Embedded Systems, Computer Architecture, Machine Learning, Artificial Intelligence, Computer Vision, Applied Software Design

EXPERIENCE

Digitalization Intern

May 2025 – Aug 2025

TITAN America – Corporate Engineering

Troutville, VA

- Built a Python automation tool that generated a nested 5,000+ folder engineering asset directory from Excel source data, standardizing infrastructure for enterprise digital records
- Developed an OCR document classification pipeline with Tesseract that oriented and routed hundreds of technical engineering documents into their correct asset locations
- Defined the taxonomy, naming conventions, and metadata schema behind the asset structure, giving the AI Planner and AI Librarian projects a queryable foundation for automated document retrieval
- Partnered with the maintenance planning team on scheduled repairs, compiling documentation that bridged corporate engineering and field operations

PROJECTS

ARK Integrity Systems | TPM 2.0, ECDSA, Embedded Linux, CUDA/TensorRT

Mar 2026 – Present

- Building a video integrity platform that cryptographically signs CCTV footage at capture using EAL4+ certified TPM 2.0 hardware key storage and ECDSA P-256, producing forensic records built for courtroom scrutiny
- Developing embedded Linux firmware for an Nvidia Jetson Orin Nano bridge that retrofits existing IP cameras: RTSP stream interception, Merkle tree frame hashing, TPM backed signing across multiple streams, hardware NVDEC decode, and CUDA/TensorRT accelerated YOLOv8 inference
- Built an open source verification tool that re-derives Merkle roots and validates ECDSA signatures and RFC 3161 timestamps, letting any attorney, court, or adjuster independently confirm footage authenticity

Orchestra | Python, FastAPI, Anthropic API, Ollama, MCP, Async/SSE

Jun 2026 – Present

- Built and open sourced a self-hosted multi-agent orchestration platform that decomposes a request into a DAG of subtasks and runs independent branches in parallel across local (Ollama) and hosted (Anthropic, OpenRouter, OpenAI compatible) models in a single run
- Designed a privacy layer enforced at a single guarded provider registry: PII redacted to stable placeholders before crossing any hosted boundary, local-only agent pinning enforced at the API, and a per-run ledger of what was sent, redacted, or blocked
- Implemented per-provider concurrency lanes so RAM-bound local inference never blocks cloud agents, cutting an 11 task run from 78.2s sequential to 43.3s wall clock

Machine Learning Pathogen Classification | PyTorch, XGBoost, scikit-learn, pandas

Jan 2026 – May 2026

- Built a team-based ML pipeline for multi-class pathogen classification across 13 public GEO transcriptomic datasets and more than 1,700 clinical patient samples
- Developed GEO data ingestion, iPAGE feature engineering, LASSO-based feature selection, and leave-one-dataset-out validation workflows for robust cross-cohort evaluation
- Trained and evaluated PyTorch neural networks, XGBoost, Random Forest, and Logistic Regression models to classify bacterial, viral, and healthy samples, reaching macro F1 above 0.53 and AUROC above 0.94 in the strongest cohorts

TECHNICAL SKILLS

Programming: Python, C, C++, Java, JavaScript, Verilog

ML/Data: PyTorch, scikit-learn, XGBoost, pandas, NumPy, Tesseract OCR

Systems/Tools: Linux, Git, FastAPI, MCP, Ollama, MATLAB, Quartus, Arduino IDE, VS Code

Hardware: Raspberry Pi CM4, TPM 2.0, Arduino, PCB Prototyping, Breadboard Circuit Design, DE-10 Lite FPGA Board, NFC/RFID, Embedded System Prototyping